Fiscal Year:	FY 2014	**Task Last Updated:**	FY 04/07/2014
PI Name:	Nelson, Gregory A. Ph.D.		
Project Title:	NSCOR: Charged Particle Radiation and Resultant Oxidative Stress Elicit Deleterious Functional Changes in the Central Nervous System		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	(1) **BMed**:Risk of Adverse Cognitive or Behavioral Conditions and Psychiatric Disorders		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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PI Organization Type:	UNIVERSITY	**Phone:**	909-558-8364
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City:	Loma Linda	**State:**	CA
Zip Code:	92350-1700	**Congressional District:**	31
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2008 NSCOR Space Radiation NNJ08ZSA003N
Start Date:	01/01/2010	**End Date:**	12/31/2014
No. of Post Docs:	9	**No. of PhD Degrees:**	
No. of PhD Candidates:	3	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA JSC
Contact Monitor:	Simonsen, Lisa	**Contact Phone:**	
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	2014 report: I. Spigelman task (UCLA) terminated. A. Obenaus tasks completed. J. Fike retired (UCSF) and tasks transferred to new CoIs S. Rosi (UCSF) and A. Allen (U. Arkansas for Med. Sciences). 2012 report: N. Komarova and D. Wodarz's tasks were not renewed for contract years 4 & 5.		
COI Name (Institution):	Soltesz, Ivan (University of California, Irvine) Limoli, Charles (University of California, Irvine) Vlkolinsky, Roman (Loma Linda University) Fike, John (Retired/ University of California, San Francisco) Spigelman, Igor (UCLA School of Dentistry) Raber, Jacob (Oregon Health & Science University) Rosi, Susanna (University of California San Francisco) Allen, Antino R (University of Arkansas for Medical Sciences)		
Grant/Contract No.:	NNX10AD59G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The proposed project will investigate responses of the mouse brain to several radiation exposure regimes corresponding to galactic cosmic rays and solar particle events. The project will characterize behavioral, electrophysiological, biochemical, and structural changes both in vivo and in vitro. Radiation-elicited oxidative stress mechanisms will be investigated as important mediators of the biological responses in both wild type and mutant animals. In addition to characterization of normal tissue responses, the potential enhancement of three late risks by radiation exposure will be explored. These are seizures, ischemic injury and traumatic injury. Mathematical models of the neurotoxicity responses will developed for use in estimating risks and fitted to project-specific data and previously-published observations. The aims of the proposal will address HRP risk 1.2 and at least five of the specific gap questions identified by NASA: CNS-1, -3, -5, -6, and -11.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Understanding the responses of the central nervous system to charged particle radiation has two main benefits – risk assessment and optimization of radiotherapy. The general population and aviation crews are exposed to charged particles during cross country and trans-oceanic air travel at high altitudes as well as to radon in their homes and workplaces. While these exposures are not as high as those of astronauts they are not insignificant. Cancer risks have traditionally been the focus of health risk estimates but the new awareness of sensitive targets in the brain such as neurogenic centers in the hippocampus raises question about the negative effects of such radiation on late neurodegenerative diseases and potential decline of cognitive abilities. This project directly addresses the cognitive and neurogenesis issues and indirectly addresses late neurodegenerative effects which share reactive oxygen chemistry with radiation as underlying mechanisms. Alzheimer's, Huntington's, and in particular, ALS diseases, are of interest. In the last ten years, the use of charged particles in radiation therapy has grown dramatically with proton and carbon ion treatment centers being established in the U.S., Europe, and Asia. Always limiting the effectiveness of radiotherapy is the response of normal tissue near the tumor site. The precise dose localization associated with charged particle therapy makes it ideal for tumors of the head and neck where careful avoidance of critical structures such as the optic nerve is critical. Knowledge of the dose "tolerance" of brain tissue and the reactions to relatively low doses is critical in optimizing these modalities of treatment.
Task Progress:	The CNS NSCOR project is proceeding on schedule and within budget incorporating recommendations of external review panels with adjustments to key personnel and contractual arrangements and plans for future experiments at Brookhaven National Laboratory and Loma Linda are underway. The electrophysiology activities at UCLA have been terminated and replaced by a UCI-led investigation of a newly-discovered microcircuit in the rodent brain to understand radiation effects on connectivity between different brain regions. The retirement of John Fike at UCSF resulted in transfer of UCSF activities to Susanna Rosi at UCSF and Antino Allen at U. Arkansas for Medical Sciences with new emphasis on the role of microglia and structural remodeling of synapses. Cells and ~900 animals have been exposed to several species of charged particles as single acute exposures and one mixed ion exposure. Data to date show complex dose response relationships and dependence on radiation quality for all endpoints examined. Persistent elevated levels of oxidative stress are observed for murine and human neural precursor cells in vitro and brain regions in vivo following irradiation. Proton, silicon, titanium and iron ions elicit cognitive, structural and electrophysiological responses at doses below 1 Gy in complex and sometimes antagonistic patterns that can be brain region specific. Novel innate immune responses have been observed after mixed ion exposure suggesting interactions between brain and peripheral monocyte compartments. The results are consistent with a role for reactive oxygen species in mediating the responses to radiation and measurements using the transgenic MCAT mouse strain seem to support our original hypothesis that enhanced catalase scavenging will ameliorate radiation-induced detrimental functional responses. Oxidative stress measurements indicate active a strong anti-oxidation response in the first few weeks post radiation with a return to baseline by one month. Computer modeling is predicting that observed membrane and synaptic properties post-irradiation can lead to altered network behavior in the hippocampus. The project is improving in integration and numerous joint manuscripts and presentations have been published by NSCOR team members and significant programmatic support has been provided by team members to the NASA HRP program.
Bibliography Type:	Description: (Last Updated: 03/26/2026)
Articles in Peer-reviewed Journals	Bezaire MJ, Soltesz I. "Quantitative assessment of CA1 local circuits: knowledge base for interneuron-pyramidal cell connectivity." Hippocampus. 2013 Sep;23(9):751-85. http://dx.doi.org/10.1002/hipo.22141 ; PubMed PMID: 23674373; PubMed Central PMCID: PMC3775914 , Sep-2013
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